

- A4) On a phase diagram, the critical temperature is _____.
 A) the temperature below which a gas cannot be liquefied
 B) the temperature above which a gas cannot be liquefied
 C) the temperature at which all three states are in equilibrium
 D) the temperature required to melt a solid
 E) the temperature required to cause sublimation of a solid

- A5) Hydration is a specific example of the phenomenon known generally as _____.
 A) salutation
 B) disordering
 C) solvation
 D) condensation
 E) dilution

- A6) As the concentration of a solute in a solution increases, the boiling point of the solution _____ and the vapor pressure of the solution _____.
 A) increases, increases
 B) increases, decreases
 C) decreases, increases
 D) decreases, decreases
 E) decreases, is unaffected

- A7) Which of the following liquids will have the lowest freezing point?
 A) aqueous CoI_2 (0.30 m)
 B) aqueous glucose (0.60 m)
 C) aqueous NaI (0.40 m)
 D) aqueous FeI_3 (0.30 m)
 E) aqueous KF (0.50 m)

- A8) For the elementary process $\text{N}_2\text{O}_5(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}_3(\text{g})$ the activation energy (E_a) and overall ΔE are 154 kJ/mol and 136 kJ/mol, respectively. What is the activation energy for the reverse reaction?
 A) 18 B) 136 C) 154 D) -18 E) 290

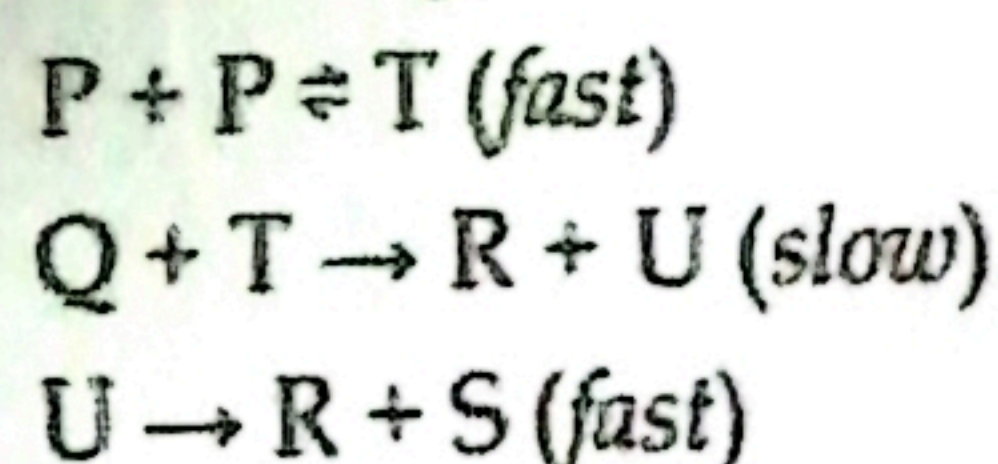
- A9) The data in the table below were obtained for the reaction:

$$\text{A} + \text{B} \rightarrow \text{P}$$

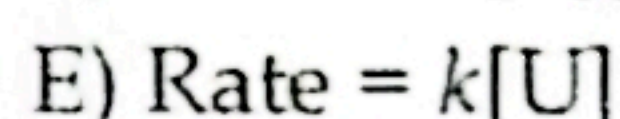
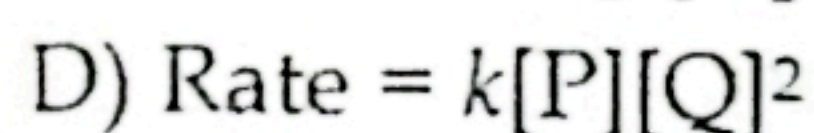
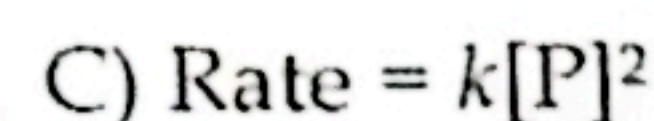
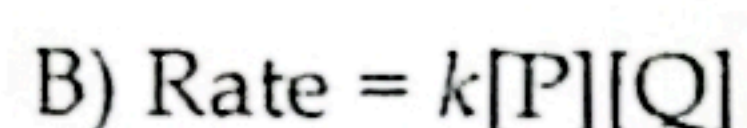
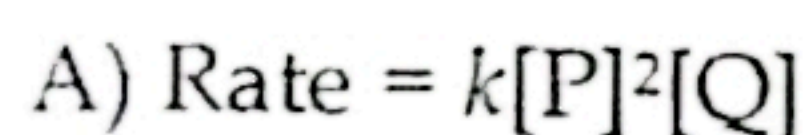
Experiment Number	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.273	0.763	2.83
2	0.273	1.526	2.83
3	0.819	0.763	25.47

- The order of the reaction in A is _____.
 A) 1 B) 2 C) 3 D) 4 E) 0

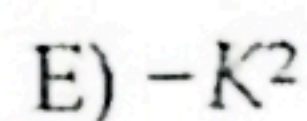
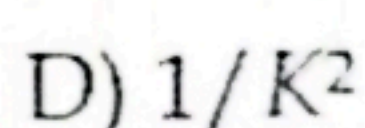
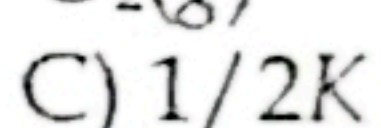
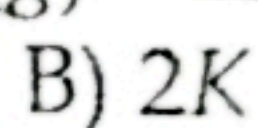
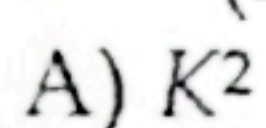
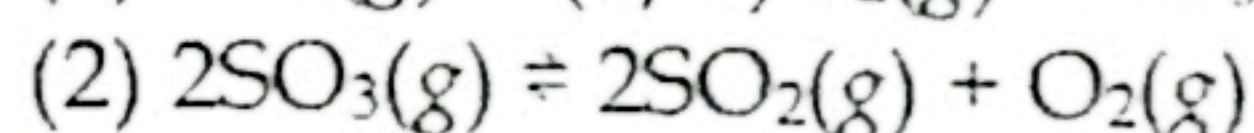
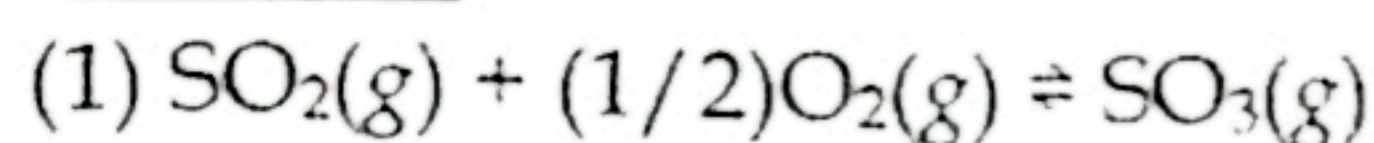
- A10) Consider the following hypothetical reaction: $2\text{P} + \text{Q} \rightarrow 2\text{R} + \text{S}$
 The following mechanism is proposed for this reaction:



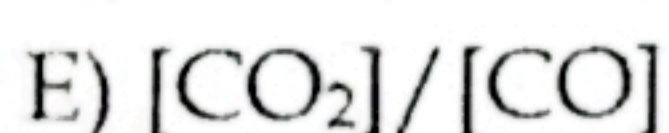
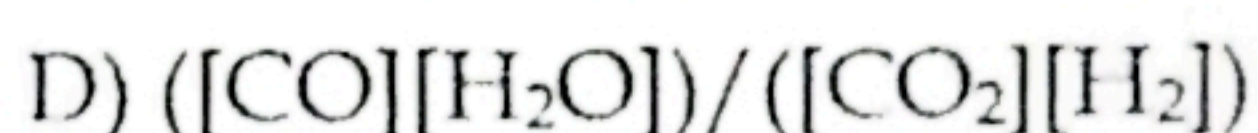
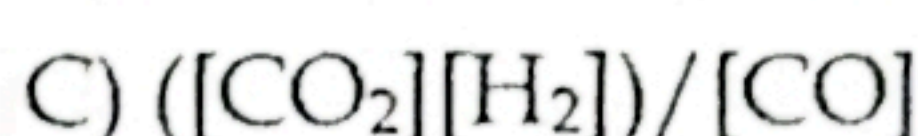
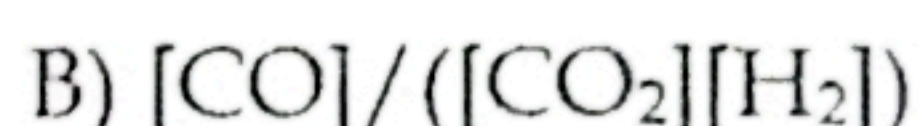
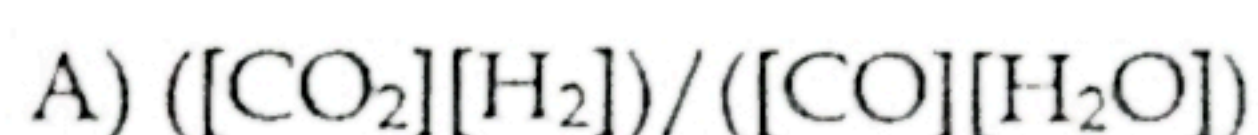
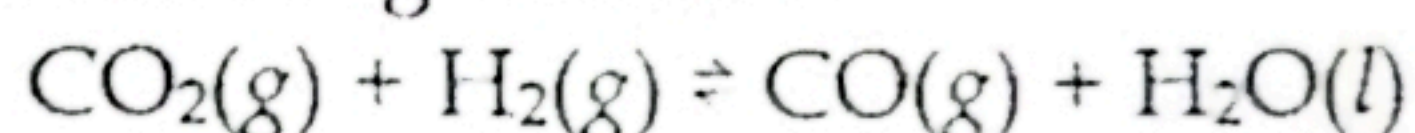
Substances T and U are unstable intermediates. What rate law is predicted by this mechanism?



A11) The equilibrium constant for reaction (1) is K . The equilibrium constant for reaction (2) is _____.



A12) Which of the following expressions is the correct equilibrium-constant expression for the following reaction?



A13) A reaction has a very large equilibrium constant of 3.3×10^{33} . Which statement is NOT true about this reaction?

A) The reaction is very fast.

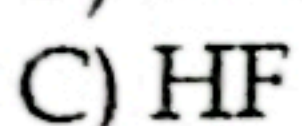
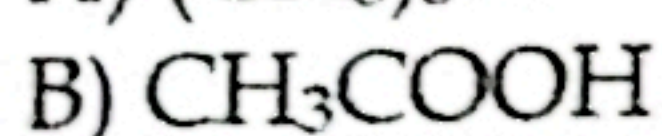
B) The reaction is essentially complete.

C) The reaction is spontaneous.

D) The equilibrium constant will change if the temperature is changed.

E) the products will react to yield very little reactant.

A14) Which one of the following is a Brønsted-Lowry base?



E) none of the above

A15) Which of the following aqueous solutions has the highest $[\text{OH}^-]$ at 25 °C?

A) a solution with a pH of 3.0

B) a $1 \times 10^{-4} \text{ M}$ solution of HNO_3

C) a solution with a pOH of 12.0

D) pure water

E) a $1 \times 10^{-3} \text{ M}$ solution of NH_4Cl

A16) Which one of the following changes is incorrect?

A) CaCl_2 added to an HCl solution does not change the pH.B) KNO_3 added to an HCl solution does not change the pH.C) CH_3COONa added to a solution of CH_3COOH raises the pH.D) NaBr added to a solution of HBr will raise the pH.E) HNO_3 added to $(\text{CH}_3\text{COO})_2\text{Ca}$ lowers the pH.

A17) What is the pH of a buffer solution that is 0.266 M in lactic acid and 0.111 M in sodium lactate? The K_a of lactic acid is 1.4×10^{-4} .

A) 14.38

B) 10.53

C) 5.38

D) 3.47

E) 4.23

A18) A reversible process is one that _____.

- A) can be reversed with no net change in either system or surroundings
- B) happens spontaneously
- C) is spontaneous in both directions
- D) must be carried out at low temperature
- E) must be carried out at high temperature

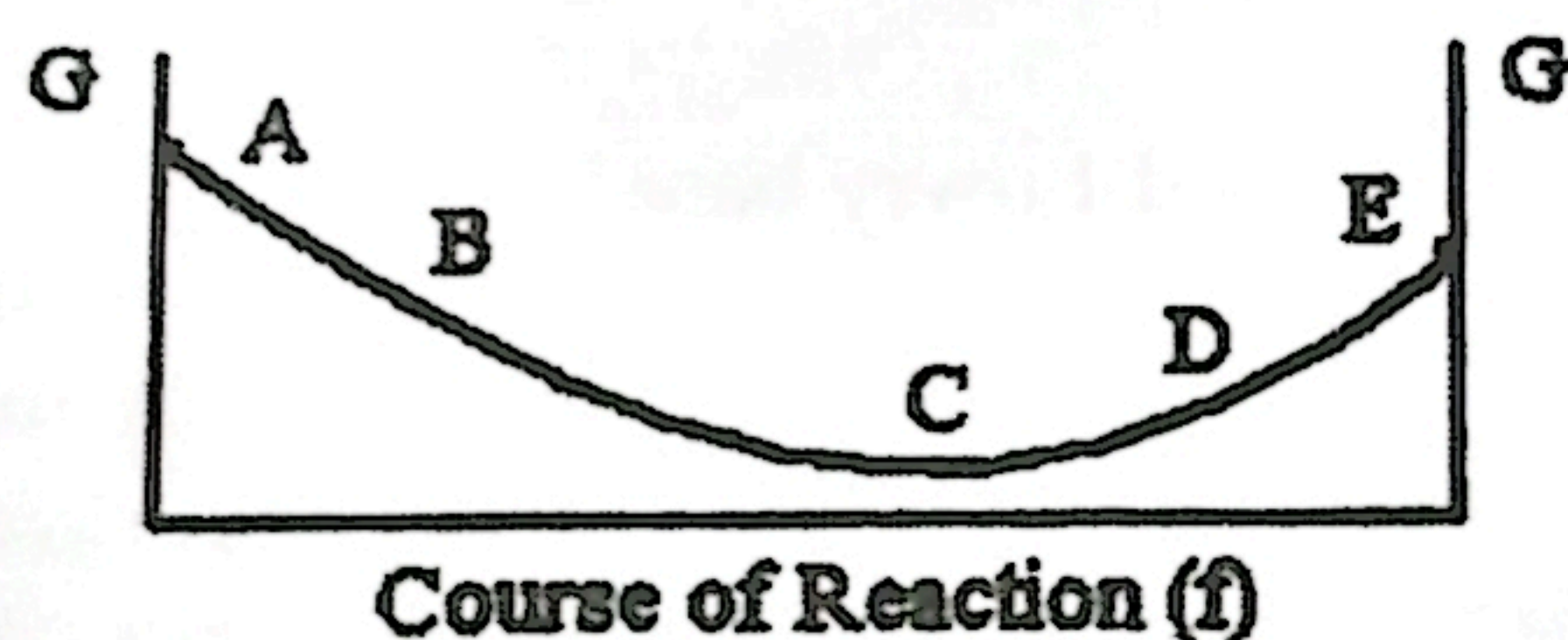
A19) Which reaction produces a decrease in the entropy of the system?

- A) $4\text{NH}_3(g) + 5\text{O}_2(g) \rightarrow 4\text{NO}(g) + 6\text{H}_2\text{O}(g)$
- B) $\text{Na}(s) + 1/2\text{Cl}_2(g) \rightarrow \text{NaCl}(s)$
- C) $2\text{HgO}(s) \rightarrow 2\text{Hg}(l) + \text{O}_2(g)$
- D) $\text{UF}_6(s) \rightarrow \text{U}(s) + 3\text{F}_2(g)$
- E) $\text{H}_2\text{O}(s) \rightarrow \text{H}_2\text{O}(g)$

A20) Which statement is false?

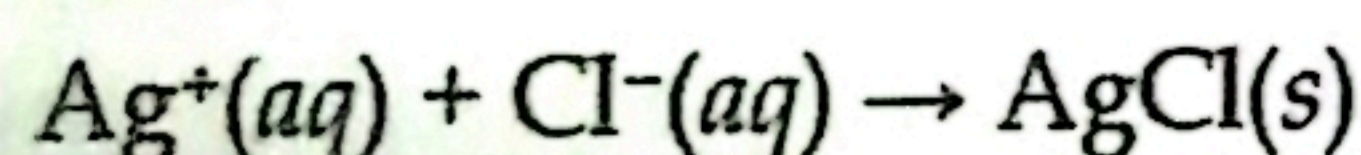
- A) ΔS for an isothermal process depends on both the temperature and the amount of heat reversibly transferred.
- B) ΔS is a state function.
- C) The second law of thermodynamics says that the entropy of the system increases for all spontaneous processes.
- D) When a system is at equilibrium, the process is not spontaneous in either direction.
- E) If the entropy of the system increases during a reversible process, the entropy change of the surroundings must decrease by the same amount.

A21) Which letter on the graph of G vs. f (course of reaction) below corresponds to the equilibrium position?



- A) A
- B) B
- C) C
- D) D
- E) E

A22) Consider the reaction:



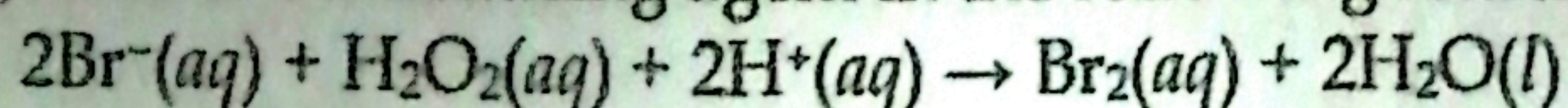
Given the following table of thermodynamic data,

Substance	ΔH_f° (kJ/mol)	S° (J/mol · K)
$\text{Ag}^+(aq)$	105.90	73.93
$\text{Cl}^-(aq)$	-167.2	56.5
$\text{AgCl}(s)$	-127.0	96.11

determine the temperature (in °C) above which the reaction is nonspontaneous under standard conditions.

- A) 1230
- B) 150
- C) 432
- D) 133
- E) 1641

A23) What is the reducing agent in the following reaction?

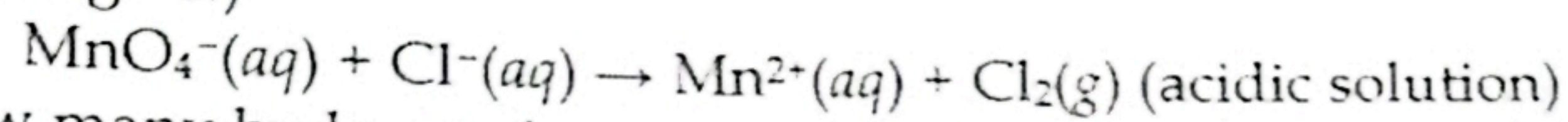


- A) $\text{Br}^-(aq)$
- B) $\text{H}_2\text{O}_2(aq)$
- C) $\text{H}^+(aq)$
- D) $\text{Br}_2(aq)$
- E) $\text{Na}^+(aq)$

A24) For a reaction that is spontaneous at standard conditions _____

- A) $\Delta G^\circ = 0$, $E^\circ = 0$, and $K > 1$
- B) $\Delta G^\circ > 0$, $E^\circ < 0$, and $K < 1$
- C) $\Delta G^\circ > 0$, $E^\circ < 0$, and $K > 1$
- D) $\Delta G^\circ < 0$, $E^\circ > 0$, and $K > 1$
- E) $\Delta G^\circ < 0$, $E^\circ = 0$, and $K \gg 1$

A25) Complete and balance the following equation. (All stoichiometric coefficients must be integers.)



How many hydrogen ions are needed and on which side of the equation must they appear?

- A) 8, on the left
- B) 8, on the right
- C) 16, on the left
- D) 16, on the right

Part B: Short Answer —Write legibly and show all work for all steps in the problem. (24 Pts)

Number of questions: 4

Pts per question: 6

B1) (a) Indicate the principal type of solute-solvent interaction in each of the following solutions: (i) KCl in water, (ii) CCl_4 in benzene (C_6H_6), (iii) methanol (CH_3OH) in water. (b) Rank the solutions from weakest to strongest solute-solvent interaction in part (a).

B2) Consider the reaction $4\text{NH}_3(g) + 5\text{O}_2(g) \rightleftharpoons 4\text{NO}(g) + 6\text{H}_2\text{O}(g)$, $\Delta H = -904.4 \text{ kJ}$. Does each of the following increase, decrease, or leave unchanged the yield of NO at equilibrium? (a) increase $[\text{NH}_3]$; (b) increase the volume of the container; (c) add a catalyst; (d) increase temperature.

B3) Select the stronger base in each of the following pairs: (a) BrO^- or ClO^- , (b) BrO^- or BrO_2^- , (c) HPO_4^{2-} or H_2PO_4^- , (d) HCO_3^- or CO_3^{2-} .

B4) (a) In a particular spontaneous process the entropy of the system decreases. What can you conclude about the sign and magnitude of ΔS_{surr} compared with ΔS_{sys} ? (b) Is it a reversible or an irreversible process in part (a)? (c) During a certain reversible process, the surroundings undergo an entropy change, $\Delta S_{\text{surr}} = -78 \text{ J/K}$. What is the entropy change of the system for this process?

B5) As the intermolecular attractive forces between molecules increase in magnitude, do you expect each of the following to increase or decrease in magnitude? (a) Vapor pressure, (b) heat of vaporization, (c) boiling point, (d) surface tension.

of lactose ($C_{12}H_{22}O_{11}$) to 200.0 g of water at 338 K. (vapor pressure of water at 338 K). (b) Calculate the concentration of CO_2 in a soft drink that is bottled with a partial pressure of CO_2 of 4.5 atm over the liquid at 25 °C. The Henry's law constant for CO_2 in water at this temperature is 3.4×10^{-2} mol/L-atm.

Part C: Long Answer -- Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (26 Pts)

Number of questions: 3

C1 (9 Pts); C2 (9 Pts); C3 (8 Pts)

C1) The reaction between ethyl iodide and hydroxide ion in ethanol (C_2H_5OH) solution, $C_2H_5I(alc) + OH^-(alc) \rightarrow C_2H_5OH(l) + I^-(alc)$, has an activation energy of 86.8 kJ/mol and a frequency factor of $2.10 \times 10^{11} M^{-1} s^{-1}$. (a) Predict the rate constant for the reaction at 35 °C with proper unit. (b) A solution of KOH in ethanol is made up by dissolving 0.335 g KOH in ethanol to form 250.0 mL of solution. Similarly, 1.453 g of C_2H_5I is dissolved in ethanol to form 250.0 mL of solution. Equal volumes of the two solutions are mixed. Assuming the reaction is first order in each reactant, what is the initial rate at 35 °C? (c) Which reagent in the reaction is limiting, assuming the reaction proceeds to completion? (d) Assuming the frequency factor and activation energy do not change as a function of temperature, calculate the rate constant for the reaction at 50 °C.

C2) The solubility of $CaCO_3$ is pH dependent at 25 °C. (a) Calculate the molar solubility of $CaCO_3$ ($K_{sp} = 4.5 \times 10^{-9}$) neglecting the acid-base character of the carbonate ion. (b) Use the K_b expression for the CO_3^{2-} ion (H_2CO_3 ; $K_{a1} = 4.3 \times 10^{-7}$, $K_{a2} = 5.6 \times 10^{-11}$) to determine the equilibrium constant for the reaction $CaCO_3(s) + H_2O(l) \rightleftharpoons Ca^{2+}(aq) + HCO_3^-(aq) + OH^-(aq)$ (c) What is the molar solubility of $CaCO_3$ at the pH of the ocean (8.3)? (d) Predicate the effect of pH on the solubility of $CaCO_3$.

C3) (a) If the equilibrium constant for a two-electron redox reaction at 298 K is 1.5×10^{-4} , calculate the corresponding ΔG° and E°_{red} . (b) $E^\circ_{red}(Ni^{2+}/Ni) = -0.760$ V, what is the pH of the solution in the cathode half-cell of a Ni- H^+ cell, when $P(H_2) = 1.00$ atm, $[Ni^{2+}] = 0.100$ M, and the cell emf is 0.542 V at 25 °C?